KAIZEN MATHS

Lesson starter

Correlation and Regression

Correlation

Name _____

SA1

For a bivariate data set, the product moment correlation coefficient is r = -0.43. Describe the correlation and state one limitation of this conclusion.

Show your working

KAIZEN MATHS

Lesson starter

Correlation and Regression

Correlation

Name _____

SB1

The table shows temperature (degrees C) and hot drinks sold (cups) for 6 observations. Calculate the product moment correlation coefficient and interpret it.

Observation	x	y	x^2	xy
1	8	101	64	808
2	11	84	121	924
3	14	75	196	1050
4	17	59	289	1003
5	20	52	400	1040
6	23	37	529	851
Total	93	408	1599	5676

Show your working

KAIZEN MATHS

Lesson starter

Correlation and Regression

Correlation

Name _____

SC1

The table shows age of car (years) and resale value (thousands of pounds) for 6 observations. The correlation coefficient is r = -0.961. Interpret this value in context.

Observation	x	y
1	1	23
2	2	19
3	3	18
4	4	14
5	5	15
6	6	11

Show your working

KAIZEN MATHS

Lesson starter

Correlation and Regression

Correlation

Name _____

SD1

For a bivariate data set, the product moment correlation coefficient is r = -0.96. Describe the correlation and state one limitation of this conclusion.

Show your working

KAIZEN MATHS

Exit ticket

Correlation and Regression

Correlation

Name _____

EA1

The table shows delivery distance (km) and delivery time (minutes) for 6 observations. Calculate the product moment correlation coefficient and interpret it.

Observation	x	y	x^2	xy
1	2	17	4	34
2	4	26	16	104
3	6	28	36	168
4	8	37	64	296
5	10	45	100	450
6	12	47	144	564
Total	42	200	364	1616

Show your working

KAIZEN MATHS

Exit ticket

Correlation and Regression

Correlation

Name _____

EB1

The table shows revision time (hours) and test score (marks) for 6 observations. The correlation coefficient is r = 0.995. Interpret this value in context.

Observation	x	y
1	1	39
2	2	48
3	3	52
4	4	61
5	5	65
6	6	74

Show your working

KAIZEN MATHS

Exit ticket

Correlation and Regression

Correlation

Name _____

EC1

For a bivariate data set, the product moment correlation coefficient is r = 0.35. Describe the correlation and state one limitation of this conclusion.

Show your working

KAIZEN MATHS

Exit ticket

Correlation and Regression

Correlation

Name _____

EA1

The table shows delivery distance (km) and delivery time (minutes) for 6 observations. Calculate the product moment correlation coefficient and interpret it.

Observation	x	y	x^2	xy
1	2	17	4	34
2	4	26	16	104
3	6	28	36	168
4	8	37	64	296
5	10	45	100	450
6	12	47	144	564
Total	42	200	364	1616

Show your working

Teacher answers & working

Correlation and Regression · Correlation

STA-00F2-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

For a bivariate data set, the product moment correlation coefficient is r = -0.43. Describe the correlation and state one limitation of this conclusion.

Answer:

weak negative correlation

Working:

Use the sign of r to decide the direction.

r = -0.43, so the relationship is negative.

Use the absolute value of r to judge the strength.

|r| = 0.43, so the relationship is weak negative.

State a limitation.

Correlation describes association; it does not prove causation.

SB1

The table shows temperature (degrees C) and hot drinks sold (cups) for 6 observations. Calculate the product moment correlation coefficient and interpret it.

Observation	x	y	x^2	xy
1	8	101	64	808
2	11	84	121	924
3	14	75	196	1050
4	17	59	289	1003
5	20	52	400	1040
6	23	37	529	851
Total	93	408	1599	5676

Answer:

r = -0.995

Working:

Use the totals from the completed table.

$\Sigma x = 93$, $\Sigma y = 408$, $\Sigma x^2 = 1599$, $\Sigma y^2 = 30436$, $\Sigma xy = 5676$

Calculate Sxx, Syy and Sxy.

$Sxx = 1599 - \frac{93^2}{6} = 157.500$

$Syy = 30436 - \frac{408^2}{6} = 2692.000$

$Sxy = 5676 - \frac{93 \times 408}{6} = -648.000$

Substitute into the PMCC formula.

$r = \frac{Sxy}{\sqrt{(Sxx \times Syy)}} = \frac{-648.000}{\sqrt{(157.500 \times 2692.000)}} = -0.995$

This is very strong negative correlation in context.

Teacher answers & working

Correlation and Regression · Correlation

STA-00F2-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1

The table shows age of car (years) and resale value (thousands of pounds) for 6 observations. The correlation coefficient is r = -0.961. Interpret this value in context.

Observation	x	y
1	1	23
2	2	19
3	3	18
4	4	14
5	5	15
6	6	11

Answer:

very strong negative correlation

Working:

The sign tells the direction.

r = -0.961 is negative.

The size tells the strength.

|r| = 0.961, so this is very strong negative correlation.

Put this into the context of the variables.

As age of car increases, resale value tends to decrease.

SD1

For a bivariate data set, the product moment correlation coefficient is r = -0.96. Describe the correlation and state one limitation of this conclusion.

Answer:

very strong negative correlation

Working:

Use the sign of r to decide the direction.

r = -0.96, so the relationship is negative.

Use the absolute value of r to judge the strength.

|r| = 0.96, so the relationship is very strong negative.

State a limitation.

Correlation describes association; it does not prove causation.

Teacher answers & working

Correlation and Regression · Correlation

STA-00F2-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1

The table shows delivery distance (km) and delivery time (minutes) for 6 observations. Calculate the product moment correlation coefficient and interpret it.

Observation	x	y	x^2	xy
1	2	17	4	34
2	4	26	16	104
3	6	28	36	168
4	8	37	64	296
5	10	45	100	450
6	12	47	144	564
Total	42	200	364	1616

Answer:

$r = 0.986$

Working:

Use the totals from the completed table.

$\Sigma x = 42, \Sigma y = 200, \Sigma x^2 = 364, \Sigma y^2 = 7352, \Sigma xy = 1616$

Calculate Sxx, Syy and Sxy.

$Sxx = 364 - \frac{42^2}{6} = 70.000$

$Syy = 7352 - \frac{200^2}{6} = 685.333$

$Sxy = 1616 - \frac{42 \times 200}{6} = 216.000$

Substitute into the PMCC formula.

$r = \frac{Sxy}{\sqrt{(Sxx \times Syy)}} = \frac{216.000}{\sqrt{(70.000 \times 685.333)}} = 0.986$

This is very strong positive correlation in context.

Teacher answers & working

Correlation and Regression · Correlation

STA-00F2-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1

The table shows revision time (hours) and test score (marks) for 6 observations. The correlation coefficient is r = 0.995. Interpret this value in context.

Observation	x	y
1	1	39
2	2	48
3	3	52
4	4	61
5	5	65
6	6	74

Answer:

very strong positive correlation

Working:

The sign tells the direction.

r = 0.995 is positive.

The size tells the strength.

|r| = 0.995, so this is very strong positive correlation.

Put this into the context of the variables.

As revision time increases, test score tends to increase.

EC1

For a bivariate data set, the product moment correlation coefficient is r = 0.35. Describe the correlation and state one limitation of this conclusion.

Answer:

weak positive correlation

Working:

Use the sign of r to decide the direction.

r = 0.35, so the relationship is positive.

Use the absolute value of r to judge the strength.

|r| = 0.35, so the relationship is weak positive.

State a limitation.

Correlation describes association; it does not prove causation.